「Def.

Let $E \subset X$ be a non-empty subset of a Metric space X. Define

$$\rho_E : X \to \mathbb{R} : x \mapsto \inf_{z \in E} d(x, z)$$

is called a distance from x to E.

「Prop. Let ρ_E be a distance from x to E in a Metric space X.

1). $\rho_E(x) = 0$ if and only if $x \in \overline{E}$.

2). ρ_E is Continuous uniformly.

Proof. To show the first, using the property of closure and infimum;

$$x \in \overline{E} \Leftrightarrow \forall \varepsilon > 0,\ B_\varepsilon(x) \cap E \neq \emptyset \Leftrightarrow \forall \varepsilon > 0,\ \exists y \in E \text{ such that } d(x,y) < \varepsilon \Leftrightarrow \inf_{z \in E} d(x,z) = 0.$$

To show the second assertion, observe that; given $x, y \in E$,

$$\rho_E(x) \leq d(x,z) \leq d(x,y) + d(y,z) \text{ for any } z \in E$$

Since it holds for all $z \in E$,

$$\rho_E(x) \leq d(x,y) + \rho_E(y)$$

and which implies $\rho_E(x) - \rho_E(y) \leq d(x,y)$. Similarly, $\rho_E(y) - \rho_E(x) \leq d(x,y)$, So

$$|\rho_E(x) - \rho_E(y)| \leq d(x,y) \text{ for any } x, y \in X.$$

Given $\varepsilon > 0$, take $\delta = \varepsilon$, then we obtain that

$$d(x,y) < \delta \Rightarrow |\rho_E(x) - \rho_E(y)| \leq d(x,y) < \varepsilon \qquad \boxtimes$$

⌈22. Let $A, B \subseteq X$ be non-empty disjoint closed sets in Metric space X. Show that a map

$$f : X \to \mathbb{R} ; x \mapsto \frac{\rho_A(x)}{\rho_A(x) + \rho_B(x)}$$

is continuous $f(x) = 0$ iff $x \in A$ and $f(x) = 1$ iff $x \in B$.

Proof Since there is no element in X s.t

$$\rho_A(x) + \rho_B(x) = 0$$

because it is equivalent to $\rho_A(x) = \rho_B(x) = 0$, but it implies that

$$x \in \overline{A} \cap \overline{B} = A \cap B = \emptyset$$

by assumption, which is contradiction. Therefore this f is well-defined. The continuity is clear. And, observe that

$$\frac{\rho_A(x)}{\rho_A(x) + \rho_B(x)} = 0 \text{ iff } \rho_A(x) = 0 \text{ iff } x \in A = \overline{A}.$$

and

$$\frac{\rho_A(x)}{\rho_A(x) + \rho_B(x)} = 1 \text{ iff } \rho_B(x) = 0 \text{ iff } x \in B = \overline{B}.$$

⌈Every closed subset $A \subseteq X$ is $Z(f)$ for some continuous map $f : X \to \mathbb{R}$.

Proof. If $A = X$, then put $f = 0$. Suppose that $A \subsetneq X$.

Then, there exists non-empty closed set $B \subset X$ s.t $A \cap B = \emptyset$. (For example, $B = \{b\}$, $b \notin A$)

Now, $f(x) = \dfrac{\rho_A(x)}{\rho_A(x) + \rho_B(x)}$, then $Z(f) = A$.

Moreover, observe that

$$A = f^{-1}[\{0\}] \subseteq f^{-1}[[0, \tfrac{1}{2})] \text{ and } B = f^{-1}[\{1\}] \subseteq f^{-1}[(\tfrac{1}{2}, 1]]$$

implies that A and B are covered by disjoint open sets.